A Predictive Study on Mental Health

STA 141A Final Project
Professor Horiguchi
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Contributions

Arjun: “Mental Health in Tech” [Dataset II] for RQ2 & RQ3. EDA, Dataset Cleaning, Methodology, Modeling, Results & Analysis. Wrote corresponding sections with Jack.

Alex: Conducted Linear Regression for Dataset I, checking conditions, and discovering r^2, adj r^2, ANOVA, and AIC/BIC values for the best model based on RQ1. Wrote corresponding comments and analysis related to Linear Regression. Also performed Linear Regression on the original question and dataset, but was ultimately scrapped due to poor results. Helped format parts of the report.

Jack: Figurehead of report, wrote executive summary, conclusion, methodology Part I along with Alex and Yihong, and analysis along with Alex and Yihong, formatting final report

Yihong: Completed k-means clustering and initial logistic regression models parts for Dataset I. Produced cross validation, confusion matrices, tables, plot visualizations for the associated methods. Wrote analysis of all of my part’s code and outputs for the report. Formatting.

Alberto: Contributed to logistic regression model and visuals for dataset 1. Helped with the formatting of the final report. Dataset selection and exploration.

Aiden: Conducted ANOVA, assumption checks, hypothesis testing, and logistic regression on original dataset, 'Mental Health and Lifestyle Habits (2019-2024),' to investigate whether online presence (screen time) influences the likelihood of anxiety or depression (Original RQ2). After statistically insignificant results, shifted focus to new RQ2 on new dataset. Formatting.

Executive Summary

Despite mental health awareness increasing in the past 100 years, mental illness is on the rise in America. Researchers have been conducting studies and performing data analysis to evaluate what risk factors play a role in mental illness development as a means for early intervention before it is too late. Our project aims to tackle the issue at heart by identifying psychological predictors of mental illness, identifying internal factors that bias one towards seeking mental illness relief, and exploring if employer interference and monetary benefits play a role in seeking medical attention. We performed our analysis on two datasets, both obtained from Kaggle, that gave us over 10,000 patient observations that we could use to answer these questions. Our first dataset was used to answer how psychological factors predict mental health risk, which we did by performing logistic regression and k means clustering. Our generated logistic regression models showed that depression was the best predictor of mental health risk, with a 70% accuracy in predicting the correct risk group. Our k-means clustering demonstrated that depression has an inverse relation on productivity in some clusters, and that social support does not influence productivity. Our second dataset was used to analyze our second and third research questions,

which we performed logistic regression, decision tree analysis, and random forest analysis on. We found that company's need to go beyond monetary benefits to support employees going through mental health struggles and that building a company culture of mental health acceptance is important. Furthermore, anonymity and observed consequences are seen as two very big factors that impact if an employee reaches out for help, but monetary benefits are key for an employee to seek self-treatment. Finally, the most important personal factor that predicts seeking treatment is family history, which we assume is due to the awareness of mental illness as a serious issue.

I. Introduction

I.I Background

Deteriorating mental health remains an ever present problem for humankind. Although mental health issues have plagued humans for millennia, it wasn't until World War I when doctors in America became concerned with the welfare of soldiers returning from war. Depressive disorders, or simply depression, are characterized by prolonged sadness and general apathy for life's activities. Along with mental health risk, depression is of interest to many researchers who wish to crack down on the causes behind their development as well as understanding what relationships exist between these disorders and personal factors.

I.II Motivation & Objectives

In this time of need to understand more about how mental health issues manifests itself into ordinary people, we have decided to take up the task of establishing what are some predictors of mental health disorders. We feel this is an extremely pressing issue as the percentage of adults identifying with mental health disorders has been on a steady incline from 2008 onwards. In this project, we hope to understand what predictors have an effect on mental illness development and how explanatory these predictors are.

I.III Research Questions

We have three main research questions we hope to answer in this project:
1) How do factors such as anxiety, social support, productivity contribute to depression, and what factors should we consider to accurately identify at-risk individuals or groups that are willing to seek help?
2) How do employer benefits and attention to mental health impact the likelihood of an employee seeking required mental attention? Specifically, are monetary benefits (remote work, work in tech, benefits, care options, wellness programs, leave) more conductive to encouraging seeking help or are cultural benefits (can ask for help, anonymity, mental health consequences, coworkers, supervisor, mental vs physical, observed consequence) ultimately more useful?
3) How impactful are genetic/physical features (age, gender, family_history) in determining the likelihood of seeking mental help?

II. Datasets

II.I Sources & Background

We obtained both of our datasets from Kaggle, a data science platform that houses many public data sets. We stumbled upon multiple datasets regarding mental health performance and prediction, but we narrowed our search down to two particular datasets. The first dataset, titled "Mental Health" and uploaded by Mahdi Mashayekhi, contains 10,000 anonymous individual observations that, although synthetically generated, are modeled on patterns in real workplace mental health datasets. The second dataset, titled "Mental Health in Tech Survey" and uploaded by Open Sourcing Mental Illness, LTD, contains 1259 data points from real workers in the tech workplace from 2014. We decided to supplement our first dataset with the second because although the data is insightful into our research questions, initial exploratory data analysis showed us that the data appeared entirely random. In order to counteract this and gain more insights into our research question, we recruited a non-synthetic dataset that looked a lot more promising in our exploratory data analysis.

II.II Predictor & Response Variables

The "Mental Health" dataset features 14 variables: Age, Gender, Employment status, Work environment, Presence of mental illness history, Whether or not the individual seeks treatment, Level of stress, Hours of sleep per night, Physical activity per week, Depression score, Anxiety score, Social support score, Productivity score, and Risk of mental health.

The "Mental Health in Tech Survey" dataset features 25 variables across personal descriptors (e.g. age, gender, family history of mental illness), company outlook regarding mental health (e.g. mental health programs, approachable coworkers, etc.), company benefits regarding mental health (e.g. programs, paid time off, etc.), and some extraneous survey questions (e.g. comments, interview culture).

III. Methodology

In order to understand how our predictors impact the response, mental health risk, we first conduct basic exploratory data analysis to unearth any underlying patterns between features and support our research questions.

Mental Health [Dataset I]

```r
##Taking a quick peek into the Mental Health Dataset:
summary(Mental_Health_Lifestyle_Dataset)
```

```
##       age           gender          employment_status  work_environment
##  Min.   :18.00   Length:10000       Length:10000       Length:10000
##  1st Qu.:30.00   Class :character   Class :character   Class :character
##  Median :41.50   Mode  :character   Mode  :character   Mode  :character
##  Mean   :41.56
##  3rd Qu.:53.00
##  Max.   :65.00
```

```
##  mental_health_history seeks_treatment     stress_level     sleep_hours
##  Length:10000          Length:10000       Min.   : 1.000   Min.   : 3.000
##  Class :character      Class :character   1st Qu.: 3.000   1st Qu.: 5.500
##  Mode  :character      Mode  :character   Median : 6.000   Median : 6.500
##                                           Mean   : 5.572   Mean   : 6.473
##                                           3rd Qu.: 8.000   3rd Qu.: 7.500
##                                           Max.   :10.000   Max.   :10.000
##  physical_activity_days depression_score anxiety_score   social_support_score
##  Min.   :0.000          Min.   : 0.00    Min.   : 0.00   Min.   :  0.00
##  1st Qu.:2.000          1st Qu.: 7.00    1st Qu.: 5.00   1st Qu.: 25.00
##  Median :4.000          Median :15.00    Median :11.00   Median : 50.00
##  Mean   :3.506          Mean   :15.04    Mean   :10.56   Mean   : 50.12
##  3rd Qu.:5.000          3rd Qu.:23.00    3rd Qu.:16.00   3rd Qu.: 76.00
##  Max.   :7.000          Max.   :30.00    Max.   :21.00   Max.   :100.00
##  productivity_score mental_health_risk
##  Min.   : 42.80     Length:10000
##  1st Qu.: 65.80     Class :character
##  Median : 77.60     Mode  :character
##  Mean   : 77.31
##  3rd Qu.: 89.20
##  Max.   :100.00
```

When exploring how variables affect one another, the most common approach to solving this problem is by using the Supervised Learning method, Linear Regression. In our question, we will be exploring whether or not we can identify a relationship between 3 predictors, and a response variable. For this specific question, we will employ linear regression modeling to see the correlation between social support, anxiety, and productivity scores as good predictors(x) to depression(y). Using tools such as linear/multiple linear regression, R^2, adjusted R^2, ANOVA, and AIC/BIC, we will be able to make our choice for which of the following combination of predictors is best fit for our data.

Before running the tests of Linear Regression, we need to first create and build the models that encompasses all the variables that we are testing for, with all different combinations as subsets. This process is known as Training the Models.

Let's train the models

```r
# full model
full_model <- lm(depression_score ~ social_support_score + productivity_score +
                   anxiety_score, data = Mental_Health_Lifestyle_Dataset)

# subsets
model_social_support <- lm(depression_score ~ social_support_score, data =
                             Mental_Health_Lifestyle_Dataset)

model_productivity <- lm(depression_score ~ productivity_score, data =
                           Mental_Health_Lifestyle_Dataset)

model_anxiety <- lm(depression_score ~ anxiety_score,
                    data = Mental_Health_Lifestyle_Dataset)
```

```r
model_social_support_productivity <- lm(depression_score ~ social_support_score +
                                          productivity_score, data =
                                          Mental_Health_Lifestyle_Dataset)

model_social_support_anxiety <- lm(depression_score ~ social_support_score + anxiety_score,
                                   data = Mental_Health_Lifestyle_Dataset)

model_productivity_anxiety <- lm(depression_score ~ productivity_score +
                                   anxiety_score, data = Mental_Health_Lifestyle_Dataset)
```

To utilize linear regression, we had to meet five assumptions: 1) our data is linear, 2) error terms are uncorrelated, 3) error terms have a constant variance (homoscedasticity), 4) no outliers, and 5) our variables are not collinear.

Validate Assumptions for Linear Regression

1. Linearity

```r
##1) Is our data Linear?

##Graphs for all predictors:
Mental_Health_Lifestyle_Dataset$full_model_residuals <- resid(full_model)

p1 <- ggplot(Mental_Health_Lifestyle_Dataset, aes(x = productivity_score,
                                                   y = full_model_residuals)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red")


p2 <- ggplot(Mental_Health_Lifestyle_Dataset, aes(x = anxiety_score,
                                                   y = full_model_residuals)) + geom_point(alpha = 0.3)
  geom_hline(yintercept = 0, color = "red")

p3 <- ggplot(Mental_Health_Lifestyle_Dataset, aes(x = social_support_score, y = full_model_residuals))
  geom_hline(yintercept = 0, color = "red")
## Wait a minute, we can see a clear pattern in Productivity score, as we have a diagonal-like pattern.
```

Making our new full model without Productivity Score:

```r
new_full_model <-lm(depression_score ~ social_support_score + anxiety_score, data =
                     Mental_Health_Lifestyle_Dataset)
Mental_Health_Lifestyle_Dataset$new_full_model_residuals <- resid(new_full_model)
```

1. Re-checking Linearity with new residuals:

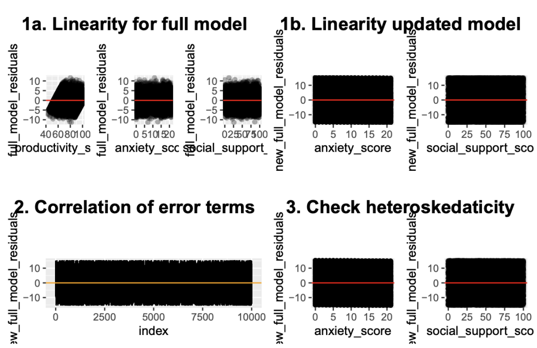
```r
p4 <- ggplot(Mental_Health_Lifestyle_Dataset, aes(x = anxiety_score,
                                                   y = new_full_model_residuals)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red")

p5 <- ggplot(Mental_Health_Lifestyle_Dataset,
             aes(x = social_support_score, y = new_full_model_residuals)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red")
```

2. No correlation between error terms

```r
##2)  Correlation of error terms:

##Let's check if our residuals are correlated over a time series:
Mental_Health_Lifestyle_Dataset$index <- 1:nrow(Mental_Health_Lifestyle_Dataset)

p6 <- ggplot(Mental_Health_Lifestyle_Dataset,
             aes(x = index, y = new_full_model_residuals)) +
  geom_line() +
  geom_hline(yintercept = 0, color = "orange")
```

3. Homoskedasticity

```r
##3)  Non-constant variances of the error terms(Heteroskedasticity)

###Refering back to the residual plots here:

p7 <- ggplot(Mental_Health_Lifestyle_Dataset,
             aes(x = anxiety_score, y = new_full_model_residuals)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red")

p8 <- ggplot(Mental_Health_Lifestyle_Dataset, aes(x = social_support_score, y =
                                                   new_full_model_residuals)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red")

#We are looking for a funnel-like shape in any of our graphs, and judging by
# the shape of all of our data, there is non-constant variance.

# Print out all the plots in one image
group_a <- arrangeGrob(
  textGrob("1a. Linearity for full model", gp = gpar(fontsize = 16,
                                                     fontface = "bold"), just = "center"),
  arrangeGrob(p1, p2, p3, ncol = 3),
  ncol = 1
)

group_b <- arrangeGrob(
  textGrob("1b. Linearity updated model", gp = gpar(fontsize = 16,
                                                    fontface = "bold"), just = "center"),
  arrangeGrob(p4, p5, ncol = 2),
  ncol = 1
)

group_c <- arrangeGrob(
  textGrob("2. Correlation of error terms", gp = gpar(fontsize = 16,
                                                      fontface = "bold"), just = "center"),
  arrangeGrob(p6, ncol = 1),
  ncol = 1
)

group_d <- arrangeGrob(
  textGrob("3. Check heteroskedaticity", gp = gpar(fontsize = 16,
                                                   fontface = "bold"), just = "center"),
  arrangeGrob(p7, p8, ncol = 2),
  ncol = 1
)

grid.arrange(group_a, group_b, group_c, group_d, nrow = 2)
```



5. No collinearity

```r
##5) Collinearity:
##Let's calculate the VIF:
###Full Model:
vif(new_full_model)
```

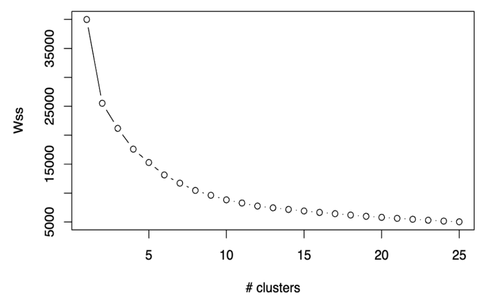
```
## social_support_score        anxiety_score
##             1.000342             1.000342
```

1. A. Linearity: Aside from Productivity Score, all other graphs show no clear pattern. Because we failed Productivity Score due to its clear pattern, we will now need to make a new full model without Productivity Score. 1. B.The new model meets the linearity assumptions. 2. Judging by the above plots, we can see that there is no clear pattern over time, as our data randomly varies. The spread is also fairly constant. 3. There is no clear pattern in any of the plots, therefore we can assume homoscedasticity. 4. We observed no outliers in our residual plots. 5. After observing all the GVIF values, we obtain values close to 1, indicating there is no serious sign of multicollinearity.

Additionally, a Logistic Regression model generates a probability of mental illness risk based on mental test scores (anxiety, depression, productivity, social support). We utilized ANOVA testing to determine whether or not the models we generated included significant predictors that have a bearing on the outcome of mental illness. Finally, we utilized k-means clustering, an unsupervised learning method, to help us understand how the data groups together into clusters. This can be useful so that we can understand how combinations of our predictor variables can correspond with different groups in our data.

```
dataForClustering <- Mental_Health_Lifestyle_Dataset[, c("depression_score",
                                                         "anxiety_score", "social_support_score",
                                                         "productivity_score")]
scaledData <- scale(dataForClustering)
```

```
# find the wss values
wss <- vector()
for (k in 1:25) {
  kmeansModel <- kmeans(scaledData, centers = k, nstart = 30)
  wss[k] <- kmeansModel$tot.withinss
}

# Plot
plot(1:25, wss, type = "b",
     xlab = "# clusters",
     ylab = "Wss",)
```



We can see from the graph that 10 is a reasonable K value, so we shall perform k-means clustering with k equal to 10.

Mental Health in Tech [Dataset II]

We took a more ground up approach by analyzing the dataset first and then generating questions based on what features provoked the most interest. The first part was cleaning and inspection.

```
# load dataset
path = Path(kagglehub.dataset_download("osmi/mental-health-in-tech-survey"))
train_df = pd.read_csv(path / "survey.csv")
```

```
# we don't need timestamp, comments, or state in our analysis
train_df.drop(columns=["Timestamp", "state", "comments"], inplace=True)

# remove the small number of missing entries
train_df.dropna(inplace=True)

# make lowercase vars for ease of use
train_df.columns = [col.lower() for col in train_df.columns]
print(train_df.columns)
print(train_df.shape)
```

```python
# statistical summaries
print(train_df.shape)
print(train_df.describe())
print(train_df.info())
```

```
(1259, 27)
                Age
count  1.259000e+03
mean   7.942815e+07
std    2.818299e+09
min   -1.726000e+03
25%    2.700000e+01
50%    3.100000e+01
75%    3.600000e+01
max    1.000000e+11
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1259 entries, 0 to 1258
Data columns (total 27 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Timestamp                  1259 non-null   object
 1   Age                        1259 non-null   int64
 2   Gender                     1259 non-null   object
 3   Country                    1259 non-null   object
 4   state                      744 non-null    object
 5   self_employed              1241 non-null   object
 6   family_history             1259 non-null   object
 7   treatment                  1259 non-null   object
 8   work_interfere             995 non-null    object
 9   no_employees               1259 non-null   object
 10  remote_work                1259 non-null   object
 11  tech_company               1259 non-null   object
 12  benefits                   1259 non-null   object
 13  care_options               1259 non-null   object
 14  wellness_program           1259 non-null   object
 15  seek_help                  1259 non-null   object
 16  anonymity                  1259 non-null   object
 17  leave                      1259 non-null   object
 18  mental_health_consequence  1259 non-null   object
 19  phys_health_consequence    1259 non-null   object
 20  coworkers                  1259 non-null   object
 21  supervisor                 1259 non-null   object
 22  mental_health_interview    1259 non-null   object
 23  phys_health_interview      1259 non-null   object
 24  mental_vs_physical         1259 non-null   object
 25  obs_consequence            1259 non-null   object
 26  comments                   164 non-null    object
dtypes: int64(1), object(26)
memory usage: 265.7+ KB
None
```

```python
# check for any missing data
nmissing_entries = train_df.isna().sum(axis=0)
print(nmissing_entries)
```

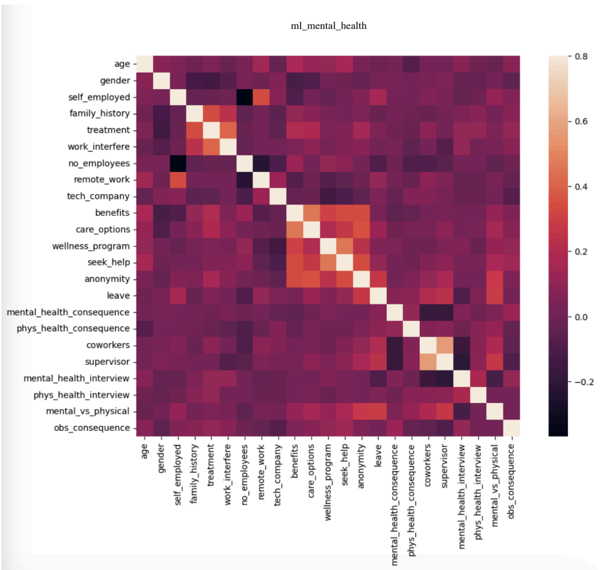
```
Timestamp                       0
Age                             0
Gender                          0
Country                         0
state                         515
self_employed                  18
family_history                  0
treatment                       0
work_interfere                264
no_employees                    0
remote_work                     0
tech_company                    0
benefits                        0
care_options                    0
wellness_program                0
seek_help                       0
anonymity                       0
leave                           0
mental_health_consequence       0
phys_health_consequence         0
coworkers                       0
supervisor                      0
mental_health_interview         0
phys_health_interview           0
mental_vs_physical              0
obs_consequence                 0
comments                     1095
dtype: int64
```

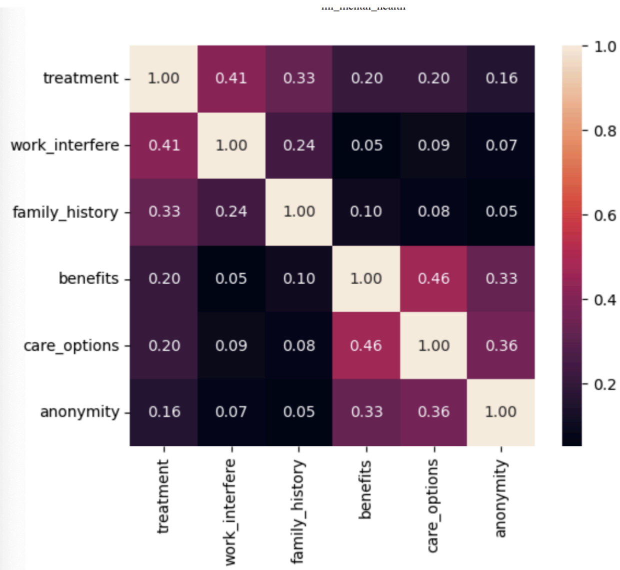
```python
other_strs = ["trans-female", "something kinda male?", "queer/she/they", "nc
female_strs = ["cis female", "f", "female", "woman",  "femake", "female ","c
remove_strs = ["nah", "p", "a little about you"]

train_df.loc[:, "gender"] = train_df.gender.str.lower().str.strip()
train_df.loc[train_df.gender.isin(male_strs), "gender"] = "male"
train_df.loc[train_df.gender.isin(other_strs), "gender"] = "other"
train_df.loc[train_df.gender.isin(female_strs), "gender"] = "female"
train_df = train_df[~train_df.gender.isin(remove_strs)]
```

```python
# remove country column prior to encoding
train_df.drop(columns=["country"], inplace=True)

# Encoding data
label_dict = {}

for feature in train_df.columns:
    # skip numeric
    if train_df[feature].dtype != "object":
        continue

    # encode the categ col
    le = LabelEncoder()
    train_df[feature] = le.fit_transform(train_df[feature].astype(str))
    label_dict[f"level_{feature}"] = list(le.classes_)

# observe mappings
for key, value in label_dict.items():
    print(key, value)
```

```python
# clean age column for neg/infinite age
train_df.loc[(train_df.age < 0) | (train_df.age > 120), "age"] = train_df.ag

# group gender column and remove any weird entries
male_strs = ["male", "m", "male-ish", "maile", "mal", "male (cis)", "make",
```

We removed out of scope (comments, state) and difficult to leverage (country) features that didn't contribute much to the report. Given we had a small number of missing entries (18), we decided to drop the rows with missing values rather than impute via median/mean. Age had some erratic values (infinite, negative) which we corrected via median imputation. We then aggregated gender into three groups (male, female, other) rather than the 20+ unique categories/variations. Then we encoded the categorical features via basic numeric encoding, i.e. assigning a unique number to each level in a given feature. This worked well since most features had some natural ordering (e.g. never, often, rarely, sometimes) that we could exploit, i.e. ordinal variables. The others were mostly binary. We then plotted correlations for all pairwise features to understand if we had to deal with multicollinearity (which we ruled out) and generated a smaller correlation matrix for the top 7 correlations with the target variable.

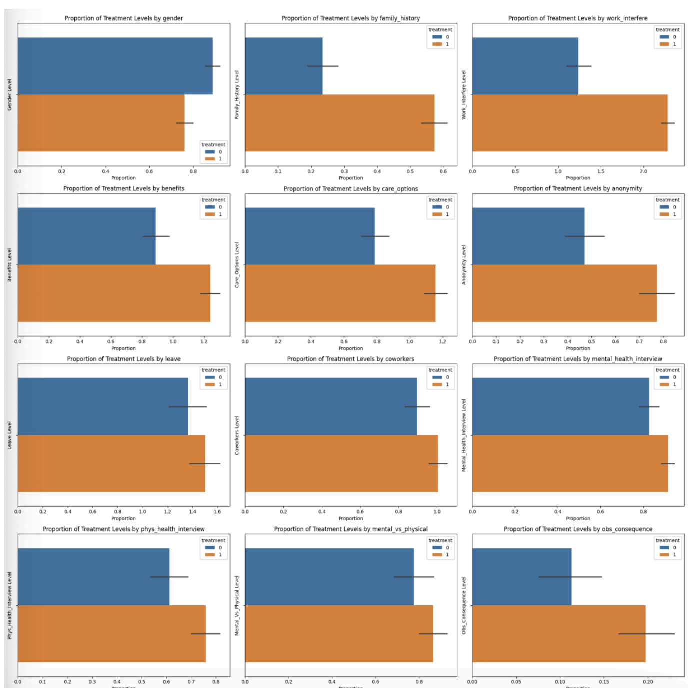




Given age was the only numeric (discrete) variable, we also wanted to visualize its distribution in greater detail. For all the categorical variables, we instead chose to plot the proportion of each level who sought treatment (i.e. P[seeks treatment | X_i = level] for all levels in feature i). We then used ANOVA to determine which categorical features met an alpha = 0.05 threshold of significance for differences in the levels and plotted them. A generated pairplot for this subset of features is also included in the Appendix.

```
# correlation matrix
corrmat = train_df.corr()
fig, ax = plt.subplots(figsize=(12, 8))
sns.heatmap(corrmat, vmax=.8, square=True)
plt.show()

# top feature corrs
k = 6
cols = corrmat.nlargest(k, "treatment")["treatment"].index
cm = np.corrcoef(train_df[cols].values.T)
hm = sns.heatmap(
    cm, cbar=True, annot=True, square=True, fmt='.2f', annot_kws={'size': 10
    yticklabels=cols.values, xticklabels=cols.values
)
plt.show()
```

```python
# iterate through categorical columns
categorical_columns = [
    "gender", "self_employed", "family_history", "work_interfere",
    "no_employees", "remote_work", "tech_company", "benefits", "care_options
    "wellness_program", "seek_help", "anonymity", "leave",
    "mental_health_consequence", "phys_health_consequence", "coworkers",
    "supervisor", "mental_health_interview", "phys_health_interview",
    "mental_vs_physical", "obs_consequence"
]

# find the columns for which a significant level difference exists
subset_cols = []
alpha = 0.05

for col in categorical_columns:
    # find the groups
    groups = [train_df[train_df[col] == level]["treatment"] for level in tra

    # ANOVA
    f_stat, p_value = f_oneway(*groups)

    # check the p_value
    if p_value < alpha:
        subset_cols.append(col)

# plot in a grid
fig, axes = plt.subplots(len(subset_cols) // 3, 3, figsize=(20, 20))
axes = axes.flatten()

for i, col in enumerate(subset_cols):
```




```python
    sns.barplot(
        data=train_df, x=col, hue="treatment", ax=axes[i]
    )
    axes[i].set_title(f"Proportion of Treatment Levels by {col}")
    axes[i].set_xlabel("Proportion")
    axes[i].set_ylabel(f"{col.title()} Level")

plt.tight_layout()
plt.show()
```

Prior to modeling, the last step was scaling the age feature to mitigate scale differences; we opted for simple Min-Max scaling given the properties of age mapped well to it.

For our model construction, we leveraged three main architectures: (1) Logistic Regression with an L2 Regularization Penalty; the most interpretable of the three, we can deconstruct its linear coefficients for an easy study of the questions we've asked. (2) Decision Tree with entropy for decision making; a more powerful architecture that can similarly be interpreted if sufficiently small. (3) Random Forests (ensemble decision trees); most powerful but least interpretable.

For each of the research questions, we fit all three models on each subset of features required and measure their performance using Accuracy, AUC (for the ROC curve), Precision, Recall. We also visualize a confusion matrix and ROC curve for all models (see appendix). We argue that a semi-well fitting model enables us to attribute importance to each of the input features, allowing us to form conclusions and reason about a company's role in promoting mental health.

IV. Analysis and Findings

Mental Health [Dataset I]

Below is the code for setting up our linear models:

```
#Current Releveant models:
new_full_model <-lm(depression_score ~ social_support_score + anxiety_score, data =
                    Mental_Health_Lifestyle_Dataset)

# individual models
model_social_support <- lm(depression_score ~ social_support_score,
                           data = Mental_Health_Lifestyle_Dataset)
model_anxiety <- lm(depression_score ~ anxiety_score, data =
                    Mental_Health_Lifestyle_Dataset)
```

```
# create the output tables for the R^2
r2_table <- data.frame(
  R_squared = sapply(all_models, function(m) summary(m)$r.squared),
  Adjusted_R_squared = sapply(all_models, function(m) summary(m)$adj.r.squared)
)

print(r2_table)
```

```
all_models <- list(
  social_support = model_social_support,
  anxiety = model_anxiety,
  all_predictors = new_full_model
)
```

```
##                    R_squared Adjusted_R_squared
## social_support 6.548028e-06      -9.347132e-05
## anxiety        1.217918e-04       2.178400e-05
## all_predictors 1.294282e-04      -7.060588e-05
```

```
anova_full_model <-anova(new_full_model)
summary(anova_full_model)
```

```
##       Df           Sum Sq            Mean Sq          F value
##  Min.   :   1   Min.   :     5.3   Min.   : 5.293   Min.   :0.06547
##  1st Qu.:   1   1st Qu.:    52.3   1st Qu.:43.067   1st Qu.:0.35625
##  Median :   1   Median :    99.3   Median :80.841   Median :0.64703
##  Mean   :3333   Mean   :269423.7   Mean   :61.818   Mean   :0.64703
##  3rd Qu.:4999   3rd Qu.:404132.9   3rd Qu.:90.081   3rd Qu.:0.93781
##  Max.   :9997   Max.   :808166.5   Max.   :99.321   Max.   :1.22859
##                                                     NA's   :1
##      Pr(>F)
##  Min.   :0.2677
##  1st Qu.:0.4003
##  Median :0.5329
##  Mean   :0.5329
##  3rd Qu.:0.6655
##  Max.   :0.7981
##  NA's   :1
```

```
AIC(model_anxiety, model_social_support, new_full_model)
```

```
##                      df      AIC
## model_anxiety         3 72306.68
## model_social_support  3 72307.83
## new_full_model        4 72308.60
```

```
BIC(model_anxiety, model_social_support, new_full_model)
```

```
##                      df      BIC
## model_anxiety         3 72328.31
## model_social_support  3 72329.46
## new_full_model        4 72337.44
```

We generated extremely low R-squared values from our regression models indicating that anxiety score and social support do not explain the variance in patient depression scores. We further evaluated our predictors significance through ANOVA testing, where we observed very large p-values and very small F values, with no significant results. Finally, we evaluated our model performance using AIC and BIC, which showed us that the anxiety model was best at predicting depression.

Although linear regression proved insignificant, we move on to logistic regression in order to understand how mental health risk can be categorized based on the predictors of depression, anxiety, social support, and productivity. We used 5-fold validation in order to find a way to validate our results from logistic regression later on.

```r
depression_model <- train(mental_health_risk ~ depression_score,
                          data=Mental_Health_Lifestyle_Dataset, method="multinom", trControl =
                            trainingSettings, trace = FALSE)

anxiety_model <- train(mental_health_risk ~ anxiety_score,
                       data=Mental_Health_Lifestyle_Dataset, method="multinom", trControl =
                         trainingSettings, trace = FALSE)

social_support_model <- train(mental_health_risk ~ social_support_score,
                              data=Mental_Health_Lifestyle_Dataset, method="multinom", trControl =
                                trainingSettings, trace = FALSE)

productivity_model <- train(mental_health_risk ~ productivity_score,
```

```r
# needed to create the same folds for both models
set.seed(121)

# Create a fold of data for the confusion matrix later
theFolds <- createFolds(Mental_Health_Lifestyle_Dataset$mental_health_risk,
                        k = 5)
testIndices <- theFolds[[1]]

trainData <- Mental_Health_Lifestyle_Dataset[-testIndices, ]
testData <- Mental_Health_Lifestyle_Dataset[testIndices, ]
testData$mental_health_risk <- factor(testData$mental_health_risk,
                                      levels = c("Low", "Medium", "High"))

# We use the remaining training folds to perform CV
trainingFolds <- createFolds(trainData$mental_health_risk, k = 5,
                             returnTrain = TRUE)

# settings for training. cross validation, 5 folds, use the predefined folds,
# and record the additional info
trainingSettings <- trainControl(method = "cv", number = 5, index = theFolds,
                                 classProbs = TRUE, summaryFunction = multiClassSummary)
```

```r
                            data=Mental_Health_Lifestyle_Dataset, method="multinom",
                            trControl = trainingSettings, trace = FALSE)

results_table <- data.frame(
  Model = c("Depression", "Anxiety", "Social Support", "Productivity"),
  Accuracy = round(c(
    depression_model$results$Accuracy[which.min(depression_model$results$logLoss)],
    anxiety_model$results$Accuracy[which.min(anxiety_model$results$logLoss)],
    social_support_model$results$Accuracy[which.min(social_support_model$results$logLoss)],
    productivity_model$results$Accuracy[which.min(productivity_model$results$logLoss)]
  ), 4),
  AUC = round(c(
    depression_model$results$AUC[which.min(depression_model$results$logLoss)],
    anxiety_model$results$AUC[which.min(anxiety_model$results$logLoss)],
    social_support_model$results$AUC[which.min(social_support_model$results$logLoss)],
    productivity_model$results$AUC[which.min(productivity_model$results$logLoss)]
  ), 4),
  Sensitivity = round(c(
    depression_model$results$Mean_Sensitivity[which.min(depression_model$results$logLoss)],
    anxiety_model$results$Mean_Sensitivity[which.min(anxiety_model$results$logLoss)],
    social_support_model$results$Mean_Sensitivity[which.min(social_support_model$results$logLoss)],
    productivity_model$results$Mean_Sensitivity[which.min(productivity_model$results$logLoss)]
  ), 4)
)

print(results_table)
```

```
##            Model Accuracy    AUC Sensitivity
## 1     Depression   0.6917 0.8525      0.6348
## 2        Anxiety   0.5927 0.7288      0.4524
## 3 Social Support   0.5892 0.4968      0.3333
## 4   Productivity   0.6681 0.8224      0.5868
```

We can see clearly that the accuracy of the models from highest to lowest is Depression -> Productivity-> Anxiety -> Social Support. Depression/productivity and social support/anxiety have similar accuracies.An interesting result is that while the anxiety and social support models have similar classification accuracies,the AUC and Sensitivity scores suggest that the anxiety model is significantly better at differentiating and identifying categories.

```r
## perform predictions on the testing data we saved earlier
depression_predictions <- predict(depression_model, newdata = testData)
anxiety_predictions <- predict(anxiety_model, newdata = testData)
social_support_predictions <- predict(social_support_model, newdata = testData)
productivity_predictions <- predict(productivity_model, newdata = testData)
# externalPredictions <- predict(external_model, newdata = testData)
```

```
## DEPRESSION MODEL
```

```r
confusionMatrix(depression_predictions, testData$mental_health_risk)$table
```

```
##            Reference
## Prediction Low Medium High
##     Low    160     95    0
##     Medium 188    921  178
##     High     0    163  295
```

```
##
## SOCIAL SUPPORT MODEL
```

```r
confusionMatrix(social_support_predictions, testData$mental_health_risk)$table
```

```
##            Reference
## Prediction  Low Medium High
##     Low       0      0    0
##     Medium  348   1179  473
##     High      0      0    0
```

```
## ANXIETY MODEL
```

```r
confusionMatrix(anxiety_predictions, testData$mental_health_risk)$table
```

```
##            Reference
## Prediction Low Medium High
##     Low    100    106    0
##     Medium 248    952  312
##     High     0    121  161
```

```
##
## PRODUCTIVITY MODEL
```

```r
confusionMatrix(productivity_predictions, testData$mental_health_risk)$table
```

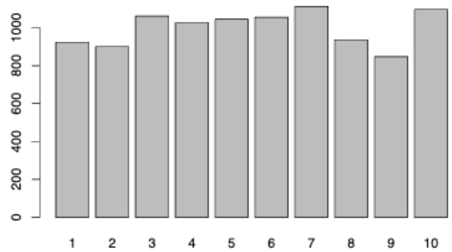
```
##            Reference
## Prediction Low Medium High
##     Low    140    106    0
##     Medium 208    932  221
##     High     0    141  252
```

With the confusion matrices of the model, the depression and productivity models appear to have better class separation, seen in the lower but spread out misclassification totals. The social support model is extremely limited because it only seems to classify medium risk for all data points. Similar to social support, the anxiety model is a bit better but also has trouble distinguishing outside of medium risk.

We finished off research question 1 by using k-means clustering with k equal to 10, with the hopes that our lack of meaningful linear correlation in the dataset could be supplemented by trends in the data.

```r
dataForClustering <- Mental_Health_Lifestyle_Dataset[, c("depression_score",
                                                         "anxiety_score", "social_support_score",
                                                         "productivity_score")]
scaledData <- scale(dataForClustering)
```

```r
kMeans <- kmeans(scaledData, centers = 10, nstart = 25)


Mental_Health_Lifestyle_Dataset$Clusters <- as.factor(kMeans$cluster)
plot(Mental_Health_Lifestyle_Dataset$Clusters)
```



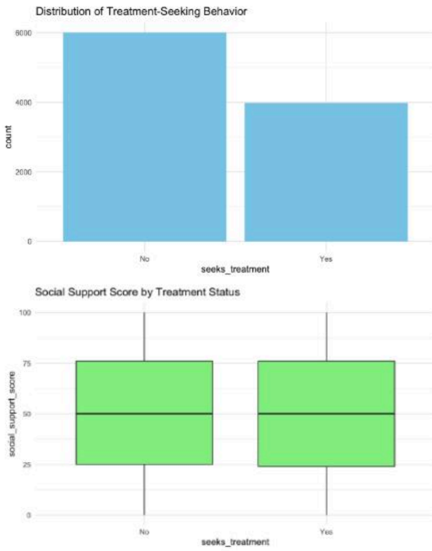

```r
clusterAverages = aggregate(
  Mental_Health_Lifestyle_Dataset[,sapply(Mental_Health_Lifestyle_Dataset, is.numeric)],
          by = list(Cluster = Mental_Health_Lifestyle_Dataset$Clusters),
          mean)
# take the average values of each cluster and print it
clusterAverages <- clusterAverages[, !(names(clusterAverages) %in%
                                          c("out_cntrl_pred_residuals",
                                            "inside_control_factors_pred_residuals", "
                                            index", "age",
                                            "stress_level", "sleep_hours",
                                            "physical_activity_days"))]
print(clusterAverages)
```

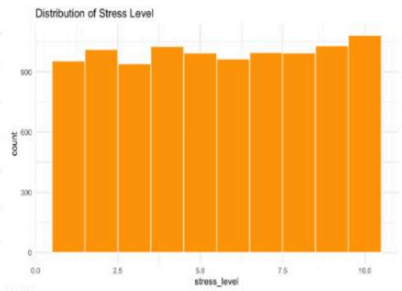
```
##    Cluster depression_score anxiety_score social_support_score
## 1        1         4.690141     15.962080             24.83424
## 2        2        25.533333      5.273333             74.10889
## 3        3         6.970782      4.789821             22.64750
## 4        4        14.589669      5.494152             73.68129
## 5        5        15.372249     15.622967             28.08995
## 6        6        22.480076      4.595825             24.51139
## 7        7        22.575540     16.245504             76.72572
## 8        8        26.077991     15.360043             25.45299
## 9        9         4.334120      5.044864             73.05549
## 10      10         7.101277     15.999088             78.21259
##    productivity_score full_model_residuals new_full_model_residuals    index
## 1            93.18028          -0.81250473              -10.4134041 5100.845
```

```
## 2            60.89178           0.62552736               10.5564140 4868.813
## 3            89.49689          -0.70115720               -7.9547635 4964.985
## 4            78.05049          -0.01487931               -0.3903685 5050.937
## 5            77.32220           0.34463970                0.2712824 4913.581
## 6            65.85152           0.60803324                7.5560040 5008.044
## 7            65.76079           0.54921813                7.4234101 4971.281
## 8            59.94936           0.62209470               10.9834121 5075.598
## 9            94.08335          -0.64059807              -10.6383013 5110.150
## 10           89.54535          -0.64350270               -8.0482312 4967.688
```

We noticed some interesting observations about our clustering. All clusters show a variation in social support score and productivity score, suggesting social support does not affect productivity. May suggest many people keep their work separate from their social life. Clusters with low depression seem to have varying low and high social support scores, suggesting that social support services may not necessarily help patients with depression. Clusters 7 and 8 both have high depression and anxiety scores, and low productivity scores. However, clusters 6 and 10 have high depression but low anxiety scores, and low depression scores with high anxiety scores. Thus suggests that depression scores should be considered more heavily than anxiety scores. All the clusters are similarly sized.

Being able to predict who is more likely to seek out the help they need can be a guide to better create intervention strategies. The use of classification methods will help understand key predictors and assess the predictive strength of a simple model.






```
Call:
glm(formula = seeks_treatment ~ stress_level + mental_health_history +
    social_support_score, family = "binomial", data = train_data)

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)            -0.3113889  0.0646582  -4.816 1.47e-06 ***
stress_level           -0.0101795  0.0079235  -1.285   0.1989
mental_health_historyYes -0.0886054  0.0500027  -1.772   0.0764 .
social_support_score   -0.0005559  0.0007847  -0.708   0.4787
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 10741  on 7999  degrees of freedom
Residual deviance: 10736  on 7996  degrees of freedom
AIC: 10744

Number of Fisher Scoring iterations: 4
```

Stress Level: The model concludes that for every unit increase in stress level there is a small decrease (-0.0101795) in log-odds of seeking treatment. However, due to the high p-value of 0.1989 this effect is not meaningful.

Mental Health History: The model concludes that having mental health history is associated with lower odds of seeking treatment (-0.0886054). With a p value of 0.0764, this is at best weak evidence.

Social Support Score: The model concludes that there is a statistically insignificant effect that comes from the social support score.

The null deviance of 10741 and the residual deviance of 10736 suggest that the current model isn't much of an improvement from the null model. The predictors explain a very small amount of variation in treatment-seeking behavior.

Mental Health in Tech [Dataset II]

With our Logistic Regression [GLM], Decision Tree [DT or Tree], and Random Forests [RF] models, we obtain the following results summarized below. Note that we refer to a pseudo research question RQ3 to gauge a baseline of utilizing all features possible, and that we zero index RQs (i.e. RQ2 corresponds to rq=1 in the tables).

The most predictively sound model seems to be RQ4 with RQ3 surprisingly not far behind, implying that although we only make use of three variables (gender, age, and family_history) for RQ3, we can sufficiently predict whether or not an employee will tend towards seeking treatment. The implication here is that the factors outside of one's control, similar to the RQ1 from Dataset I, are quite powerful indicators of whether someone will attempt to improve their mental health.

Observing RQ2 part (a) (monetary benefits) against the performance of the RQ2 part (b) models, we see a non-trivial prediction difference between the two. Specifically, the models support evidence that a company needs to go beyond monetary/work benefits into building a supportive culture. It seems that the more support employees receive from their peers and supervisors, the higher their propensity to utilize the mental health resources their company/government provides.

From a modeling perspective, it seems as though the RF models are by far the most capable predictors. Objectively, the RF architecture is the most flexible of the three models, indicating that the mental health tech survey dataset is likely more non-linear in its patterns. The GLM, however, isn't far behind RF and provides far more interpretability. Last, the Tree model seems to be the weakest of the three, although this could be due to us avoiding a computationally expensive grid search for hyperparameters (for both RF and Tree).

	0	1	2	3	4	5	6	7	8	9	10	11
model	glm	tree	rf	glm	tree	rf	glm	tree	rf	glm	tree	rf
RQ	1	1	1	1	1	1	2	2	2	3	3	3
Part	a	a	a	b	b	b	a	a	a	a	a	a
Accuracy	0.593857	0.583618	0.600683	0.62116	0.617747	0.651877	0.723549	0.720137	0.703072	0.771331	0.750853	0.795222
AUC	0.552586	0.596896	0.582461	0.547007	0.524508	0.547272	0.741725	0.730251	0.698207	0.796003	0.800153	0.826882
Precision	0.6875	0.727273	0.69802	0.679167	0.66537	0.697095	0.841176	0.804233	0.777778	0.809524	0.816327	0.804444
Recall	0.725888	0.609137	0.715736	0.827411	0.86802	0.852792	0.725888	0.771574	0.781726	0.862944	0.812183	0.918782

With these initial conclusions in mind, we move onto interpreting the model itself. As previously mentioned, the GLM and Tree offer the most interpretability of the three models. However, given the performant nature of the GLM in this setting, we opt to analyze its coefficients over the decisions in the Tree model.

```python
def match_coeffs(rq: int, part: str, subset: list[str]):
    coef_df = pd.DataFrame({
        "Feature": subset,
        "Coefficients": trained[f"RQ{rq}"][part]["glm"].coef_[0]
    })
    return coef_df

# print report for each rq part
for rq, part, subset in zip([1, 1, 2, 3], ["a", "b", "a", "a"],
chain.from_iterable(rqs)):
    print(f"Coefficients for GLM model in {rq} - Part {part}:")
    display(match_coeffs(rq=rq, part=part, subset=subset))
```

Note that we include the coefficients for the pseudo question RQ4 in the appendix. Given that we scaled all features prior to model fitting, we can leverage the model's coefficients in one-to-one

Coefficients for GLM model in 1 – Part a:

	0	1	2	3	4	5
Feature	remote_work	tech_company	benefits	care_options	wellness_program	leave
Coefficients	0.14273	-0.446687	0.539462	0.31633	-0.13024	0.114154

Coefficients for GLM model in 1 – Part b:

	0	1	2	3	4	5	6
Feature	seek_help	anonymity	mental_health_consequence	coworkers	supervisor	mental_vs_physical	obs_consequence
Coefficients	0.144246	0.451141	-0.046878	0.437376	-0.247698	-0.017157	0.66336

Coefficients for GLM model in 2 – Part a:

	0	1	2
Feature	age	gender	family_history
Coefficients	0.635601	-0.576347	1.295329

comparisons with each other for feature attribution (i.e. importance of each feature in model predictions).

For RQ2, we observe that anonymity and observed consequences play the biggest role in whether or not an employee reaches out for help. We thereby quantitatively prove that a culture that penalizes those who leverage mental health resources is harmful in encouraging those to seek aid. It seems that coworkers, more than supervisors, play a part in creating a culture of acceptance–if an employee feels supported by their peers enough to reach out to them, then they're more likely to seek professional help as well. We also note that monetary benefits do make a significant contribution to encouraging self-treatment, presumably because they enable a purely monetary decision (of seeking professional help) to become a purely emotional one.

For RQ3, we observe that family history is by far the most important factor in predicting whether an individual seeks treatment. We posit that the trauma of seeing someone else go through mental health challenges teaches individuals the importance of reaching out for help and what resources they have.

V. Conclusion

RQ1: Our results indicate that anxiety, social support, and productivity do not have a linear relationship with depression. There appears to be a more complex relationship between mental illness development and psychological predictors than we imagined before conducting this analysis. However, we did find that we could make accurate predictions based on these predictors to assign patients to a low, moderate, or high risk of developing mental illness. We found that depression alone generated the most accurate linear regression model, indicating that the level of depression a person exhibits on a depression assessment is a very strong predictor of what their mental health risk should be. Furthermore, our cluster analysis indicates that depression scores appear to have the strongest influence on productivity scores, with higher depression leading to less productivity. In regards to mental health awareness and the rise of mental illness, it appears that one of the most important risk factors among many is depression and depressive behavior. We saw from our clustering that depressed people can be socially supported and low in anxiety, so it is not as easy to make out whether somebody is depressed or not. However, a continued push for more mental check-ins with a doctor may help to identify those high in depression to prevent the festering of mental illness.

RQ2: Our results demonstrate that employer benefits alone are not conducive to seeking mental health services, but that a created culture of acceptance in the workplace can be. Creating a culture of acceptance requires a large group effort, which can only be made by further increasing awareness of mental illness concerns. Not only do managers and HR need to take initiative in this regard, but fellow coworkers must do their duty in understanding the importance of mental health. In regards to HR, anonymous conversations and between them and employees as well as security from consequence was found to be essential in employees seeking mental attention, highlighting the importance of a safe, strong company culture. It appears that if employers take care of their employees, by means of secure culture and monetary incentive, employees will utilize the healthcare system to focus on their mental health. Seeking help can be seen as a very expensive issue to fix, so by receiving financial relief from employers, employees can learn to take care of themselves.

RQ3: Our results explain that the most important personal factor in determining whether or not a person seeks mental help is the existence of family history of mental illness. We understand this result because although many people are only just becoming more aware of mental illness and its impacts, those growing up with a mentally ill family member have dealt with these issues for most of their life. These people have a lot more experience with the negative side effects of holding off on treating mental illness, and are also likely aware that some mental illness is genetic. Taking these precautions against the development of mental illness before its onset is extremely proactive and important considering the example it sets. However, this again raises the concern that those without first hand experience with mental illness may be quick to shrug off the risk factors as if it is no big deal. With the increase in mental health awareness and increase in mental illness prevalence in young adults, we hope that the seriousness of mental health is seen as more real in the coming years.

Although we discovered some meaningful insights from our project, it is clear that there were some limitations to our findings. Firstly, our original dataset was artificially generated, leading us to need to supplement our data sources with an additional one. As well as this, the additional dataset originates from a 2014 tech study, indicating that this data may be outdated or at the very least inapplicable to the entire population of the United States. We would be more likely to generate more meaningful and applicable data assuming we could access a current random representative sample of Americans, but unfortunately we were constrained to what was available on the internet. Despite this, we were able to answer some very important questions regarding mental illness prevention and intervention. Considering mental health was only made important about 100 years ago, we have leaps and bounds of progress still to make on how to best resolve our mental illness crisis.

Bibliography

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https://www.mayoclinic.org/diseases-conditions/coronavirus/in-depth/mental-health-covid-19/art-20482731

Code & All Results for “Mental Health” [Dataset I]

R Markdown File

Knitted PDF

Code & All Results for “Mental Health in Tech” [Dataset II]

Exported HTML from Notebook
Source Code (Jupyter Notebook)
Exported PDF from Notebook